

Dr. NAKUL TANWAR

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RESEARCH PROFILE

Bioinformatician with demonstrated research expertise in explainable AI for molecular disease characterization, omics-driven biomarker discovery, and reproducible computational pipelines. PhD research focused on integrating multi-modal molecular datasets to identify novel biomarkers in respiratory disorders through explainable machine learning. Strong publication record with peer-reviewed articles in high-impact journals. Experienced in designing rigorous computational workflows, mentoring junior researchers, and translating complex biological data into actionable clinical insights.

EDUCATION

Ph.D. Scholar, Department of Biotechnology

- Delhi Technological University | 2022–2026 (Awarded)
- Thesis Title: An Integrated Approach Towards the Identification of Novel Biomarkers in Respiratory Disorders
- Supervisor: Prof. Yasha Hasija
- Research Focus: Explainable AI for miRNA-mediated pathways in COPD, pulmonary fibrosis, and lung cancer; multi-omics integration; molecular mechanism elucidation

MSc Biotechnology

- Delhi Technological University | 2020–2022
- Project Title: Explaining Developmental Neurotoxicity By XAI
- Supervisor: Prof. Yasha Hasija

B.Sc. (Honors) Biotechnology

- Amity University, Gurugram | 2017–2020

RESEARCH PROJECTS (PROOF OF CONCEPT)

A. DisProtCellLLM (Disease Protein-Cell Large Language Model) | Postdoctoral Research Proposal | 2026

- Designed a three-tower multimodal deep learning framework that fuses scRNA-seq expression (Scanpy), ESM2 protein language model embeddings and disease-contextual biomedical text through two-stage cross-attention, with LLM-generated (Mistral 7B) protein-structure-informed drug target hypotheses.
- Proof of concept on small cell lung cancer scRNA-seq (GSE319155, ~26.8k cells): AUROC 0.85 for the PCA+ESM2 model under patient-level held-out evaluation (5 train / 2 test patients), versus 0.55–0.70 for transcript-only baselines.

B. SCOPE (Spatially Contextualized Omics with Protected Explanations) | Extension of DisProtCellLLM | 2026

- Extended DisProtCellLLM into a five-tower spatial omics framework (transcriptomics, ESM2 protein, pathway, spatial graph, histology morphology via ResNet50) with an LLM factuality-grounding layer that validates LLM-proposed genes against observed expression.
- Proof of concept on one ES-SCLC GeoMx DSP slide (7 ROIs, GSE261348): resolved morphologically similar regions with divergent expression, found three unsupervised microenvironment zones (immune-infiltrated, immunosuppressive stromal, proliferative tumor core), and quantified zone-dependent LLM hallucination (0% in tumor core, 30% in stroma). Exploratory; validation on 175 ROIs from 19 patients is planned.

PUBLICATIONS & RESEARCH CONTRIBUTIONS

A. Peer-Reviewed Journal Articles

- **Tanwar, N.,** Hasija, Y. (2025). "Explicating miRNA-mediated regulation of inflammatory pathways in COPD, MS, and lung cancer using explainable artificial intelligence: insights from peripheral blood profiles." Integrative Biology, 17, zyaf020. <https://doi.org/10.1093/intbio/zyaf020>
- **Tanwar, N.,** Hasija, Y. (2024). "Explicate molecular landscape of combined pulmonary fibrosis and emphysema through explainable artificial intelligence: a comprehensive analysis of ILD and COPD interactions using RNA from whole lung homogenates." Medical & Biological Engineering & Computing, 62, 2557–2570. <https://doi.org/10.1007/s11517-024-03099-8>
- **Tanwar, N.,** Hasija, Y. (2025). "An eXplainable AI and Conversational Decision Support System for Lung Cancer Diagnosis and Monitoring." Technology and Health Care. (Manuscript Under Review). <https://zenodo.org/records/15570514>

B. Book Chapter

- **Tanwar, N., & Hasija, Y. (2024).** "Overview of new trends on deep learning models for diabetes risk prediction." In Internet of Things and Machine Learning for Type I and Type II Diabetes (pp. 135-146). Elsevier.

C. IEEE Conference Presentations

- Yadav, K., **Tanwar, N., & Hasija, Y. (2024).** "Exploring the role of gut microbiota in hypertension: Insights from Machine Learning and eXplainable AI." 15th International Conference on Computing, Communication and Networking Technologies (ICCCNT), June 2024.
- **Tanwar, N., Meena, J., & Hasija, Y. (2022).** "Explicate Toxicity by eXplainable Artificial Intelligence." International Conference on Industry 4.0 Technology (I4Tech), September 2022.
- **Tanwar, N., & Hasija, Y. (2022).** "Explainable AI: Are We There Yet?" IEEE Delhi Section Conference (DELCON), February 2022.
- **Tanwar, N., & Hasija, Y. (2022).** "Deep Learning: A Tool in Biomedical Science." International Conference for Advancement in Technology (ICONAT), January 2022.

RESEARCH EXPERIENCE & TEACHING CONTRIBUTIONS

A. Teaching & Mentoring | Delhi Technological University | 2022–2025

- Mentored undergraduate and postgraduate students in bioinformatics research projects, providing guidance on experimental design, data analysis workflows, and ML model interpretation
- Co-designed and co-taught modules on explainable AI applications in biomedical sciences for B.Tech and M.Tech programs
- Supervised research documentation ensuring reproducibility, rigorous methodology, and publication-quality outputs
- Developed reproducible computational tutorials for omics data analysis and XAI model interpretation, enhancing student competency in modern bioinformatics

B. COVID-19 Molecular Diagnostics | Imperial Life Sciences | May–July 2021

- Validated RT-PCR assay kits for SARS-CoV-2 detection, optimizing reagent formulations and reaction conditions
- Established quality control protocols ensuring assay sensitivity, specificity, and reproducibility across batch production
- Contributed to rapid diagnostic tool development during pandemic response, supporting public health initiatives

C. Clinical Diagnostics & Pathology | Janakpuri Super Specialty Hospital | May–July 2019

- Performed comprehensive diagnostic analyses (CBC, blood smears, histopathology preparation)
- Mastered clinical workflow optimization, equipment maintenance, and quality assurance in diagnostic laboratory settings

CORE RESEARCH COMPETENCIES

A. Computational Omics Analysis

- Omics data integration (RNA-seq, miRNA profiling, whole lung homogenates)
- Statistical analysis, dimensionality reduction, pathway enrichment
- Hands-on expertise in differential expression analysis and molecular biomarker discover.

B. Single-Cell & Spatial Omics

- scRNA-seq analysis: QC, normalization, PCA, batch correction (Scanpy, Harmony)
- Spatial transcriptomics analysis (GeoMx DSP): ROI-level profiling and spatial zone characterization
- Patient-level, held-out evaluation to avoid data leakage across cells and patients
- Tumor microenvironment analysis in SCLC (immune-infiltrated, stromal, proliferative zones)

C. Multimodal Deep Learning & Protein Language Models

- Multi-tower architectures fusing expression, protein, text, spatial graph and histology features
- Cross-attention fusion of omics with protein embeddings (ESM2)
- Spatial graph (k-NN) and histology morphology features (ResNet50)
- Text embeddings from biomedical literature (BioMedBERT, TF-IDF + SVD)

D. Machine Learning & Explainable AI (XAI)

- Supervised and unsupervised learning model development
- Feature engineering, model evaluation, and validation strategies
- Explainability frameworks: SHAP, LIME, attention mechanisms
- Deep learning architectures for biomedical applications

E. Large Language Models & Biomedical NLP

- LLM-augmented pipelines with XAI for biomedical text interpretation

- Prompt optimization for clinical decision support
- Conversational AI systems for disease diagnosis and monitoring
- Model evaluation and reliability assessment

F. Data Science & Reproducibility

- Statistical analysis and exploratory data analysis (EDA)
- Data cleaning and quality assurance
- Reproducible research practices and workflow documentation
- Python scientific computing (NumPy, Pandas, scikit-learn, Matplotlib PyTorch, TensorFlow, Cross Attention fusion)

G. Research Methodology

- Experimental design and hypothesis formulation
- Rigorous statistical testing and result interpretation
- Critical literature review and synthesis
- Peer review and publication processes

TECHNICAL SKILLS

- **Programming Languages:** Python (scientific computing, ML, automation)
- **ML Frameworks:** scikit-learn, TensorFlow/Keras, PyTorch, XGBoost
- **Data Analysis:** Statistical testing, EDA, dimensionality reduction, feature engineering
- **Bioinformatics Tools:** DESeq2, pathway analysis, gene set enrichment
- **Single-Cell & Spatial Tools:** Scanpy, Harmony, GeoMx DSP data analysis
- **Protein & Language Models:** ESM2, BioMedBERT, Mistral 7B (Ollama)
- **Computer Vision:** ResNet50 (histology morphology features)
- **Graph Methods:** k-NN spatial graphs, cross-attention fusion
- **LLM & NLP:** LLM prompt engineering, text classification, biomedical knowledge extraction
- **Data Visualization:** Matplotlib, Seaborn, Plotly, interactive dashboards
- **Version Control & Reproducibility:** Github, Jupyter Notebooks, documentation best practices

RECOGNITION & HONORS

- **Commendable Research Award**, Delhi Technological University (Research & Innovation Excellence Awards 2026) Recognition for outstanding contributions to explainable AI research in respiratory disease characterization
- **Commendable Research Award**, Delhi Technological University (Research & Innovation Excellence Awards 2025) Recognition for outstanding contributions to explainable AI research in respiratory disease characterization
- **GIAN Course Completion:** Medical Translational Research – Bench to Bedside (2025) Instructor: Prof. Hitesh Handa. ‘Advanced coursework in translating bench research to clinical applications’

INTELLECTUAL PROPERTY

- **Design Patent (Under Consideration)**
Title: Portable Food Spoilage Detection Device – Dual-sensor (gas + TDS/pH) system
Application No.: 461910-001
Status: FER Review (2025)